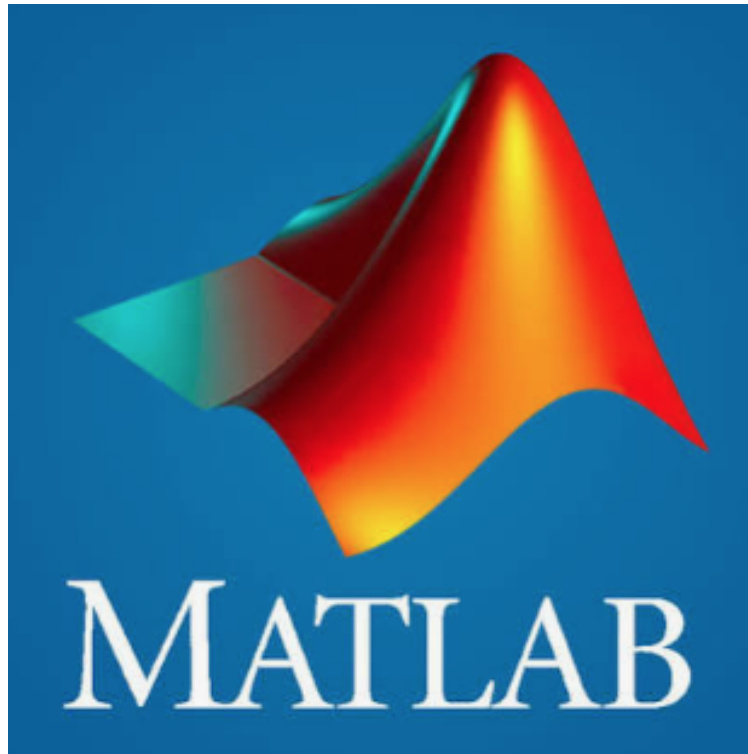
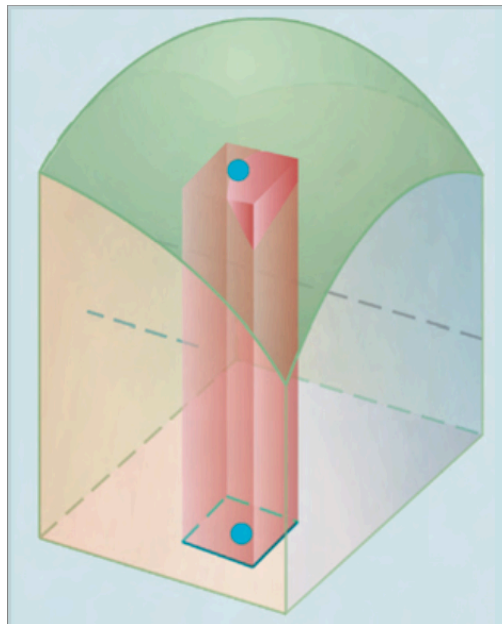


MATH 280 MATLAB Project #4 - Multiple Integral and Mass Properties Analyzer



When the project is completed, rename the end of the file to (Completed Implementation).



PROJECT OUTLINE

Core Objective

Develop a MATLAB application that allows users to visualize and evaluate double and triple integrals over various regions. The application provides interactive tools for calculating area, volume, mass, moments, and centers of mass while displaying the integration regions through an intuitive graphical user interface.

Rather than acting as a basic numerical integration calculator, the application serves as an interactive Engineering Integration and Mass Properties Analysis Tool.

Concepts Covered

Chapter 15 Multiple Integrals

15.1 Double Integrals

15.2 Area, Moments, and Centers of Mass

15.3 Double Integrals in Polar Form

15.4 Triple Integrals In Rectangular Coordinates

Learning Objectives

By completing this project you will practice:

- double integration
- triple integration
- polar integration from Calculus 2
- numerical integration
- coordinate transformations
- engineering applications
- center of mass calculations
- moment calculations
- 2D and 3D visualization

- Designing MATLAB Apps

Required Features

1. Region Visualizer in MATLAB

User enters:

- a) Limits of Integration
- b) Function
- c) Coordinate system (cartesian or polar)

MATLAB displays:

- shaded integration region
- boundary curves
- labeled axes
- coordinate grid

This feature should support common MVC regions such as rectangles, triangles, circles, ellipses, and user-defined boundaries.

2. Double Integral Calculator/Analyzer in MATLAB

MATLAB computes:

- numerical approximation
- symbolic solution (when possible)

And displays:

- integration region
- surface
- accumulated value
- The feature should also highlight the region bounded by the integral.

3. Triple Integral Calculator/Analyzer in MATLAB

User enters a 3D region

MATLAB computes:

- volume
- mass
- accumulated quantity

And displays:

- transparent solid
- slicing planes
- integration direction animation

Works very similar to the double integral feature but with additional steps.

4. Area Calculator

MATLAB displays:

- shaded region
- computed area
- comparison with geometric formulas when applicable

This feature simply computes the area of planar regions using double integrals.

5. Center of Mass and Density Explorer in MATLAB

MATLAB displays:

- object geometry
- density distribution
- center of mass location

This feature allows users to move the density distribution interactively and observe how the center of mass changes in real time.

6. Mass, Moments, and Center of Mass Calculator from Density Function in MATLAB

User enters the density function $\rho(x)$

MATLAB computes:

- total mass
- moment about x-axis
- moment about y-axis
- center of mass

And displays:

- density heat map
- center of mass marker
- mass distribution visualization

7. Polar Coordinate Explorer in MATLAB

User selects one of the following

- circle
- annulus
- sector
- cardioid
- rose curve
- custom polar region

MATLAB:

- converts the region
- computes the integral

- visualizes the transformed domain

This feature allows users to compare Cartesian and polar coordinate representations side by side.

8. Engineering Application Library in MATLAB

Include built-in demonstrations such as

- mass of a nonuniform plate
- volume of a storage tank
- density of composite materials
- heat distribution over a surface
- fluid-filled container
- probability density regions

These are common applications and all of the three advanced maths (MVC, Differential Equations, Linear Algebra) have some kind of applications.

Users can modify parameters and immediately observe changes in calculated quantities.

9. Animated Integration in MATLAB

Animate the following in MATLAB:

- sweeping integration strips
- polar sectors
- stacked volume elements
- accumulation process

Users can adjust animation speed and integration direction using this feature.

10. Numerical Method Comparison in MATLAB

MATLAB compares:

- MATLAB numerical integration
- symbolic integration (where available)
- user-selected approximation methods

Display convergence and relative error when an exact solution exists.

Necessary MATLAB Files For This Project/File Structure

MVC_Project4/

main.mlapp

main.m

doubleIntegralCalculator.m

tripleIntegralCalculator.m

polarIntegralCalculator.m

regionGenerator.m

volumeCalculator.m

massCalculator.m

momentCalculator.m

centerOfMassCalculator.m

plotUtilities.m

README.md

Useful MATLAB Commands For This Project

integral()

integral2()

integral3()

trapz()
meshgrid()
surf()
fsurf()
contour()
contour3()
fill()
fill3()
patch()
scatter3()
plot3()
quiver3()
hold on
axis equal
view()
uifigure()
uiaxes()
uibutton()
uidropdown()
uislider()
uitable()
uicontrol()

Additional MATLAB Pseudocode

Double Integral

Receive function

Receive region

Evaluate integral

Display numerical result

Visualize region

Polar Integration

Receive polar bounds

Transform coordinates

Apply Jacobian

Evaluate integral

Display transformed region

Triple Integral

Receive function

Receive volume bounds

Evaluate integral

Visualize solid region

Center of Mass

Receive density

Compute mass

Compute moments

Calculate center of mass

Display location

Volume Visualization

Generate solid

Apply transparency

Show slicing planes

Animate accumulation

GUI FUNCTIONALITY

The GUI emphasizes exploration and visualization:

- Function input for density and integrand

- Region selector (rectangle, triangle, disk, cylinder projection, custom)
- Coordinate system toggle (Cartesian (x,y,z) or Polar (r, θ) coordinates)
- Interactive sliders for changing integration bounds

Checkboxes to display:

- Integration region
- Density map
- Volume shading
- Center of mass
- Coordinate grid
- Animation controls for visualizing the integration process
- Export numerical results to a table (e.g. Excel)
- Save plots as downloadable images

MAIN MENU

Multiple Integral Analyzer

1. Plot Integration Region

2. Double Integrals

3. Area Calculator

4. Mass and Moments

5. Polar Coordinate Integrals

6. Triple Integrals

7. Center of Mass

8. Engineering Applications

9. Export Results

10. Exit

GUI LAYOUT

Primary Interface

Left Panel

Input fields

- Function $f(x,y)$ or $f(x,y,z)$
- Integration limits
- Density function
- Coordinate system
- Region type

Buttons

- Plot Region
- Integrate
- Reset

Center

Interactive visualization

- 2D regions
- 3D solids
- Polar regions

- Volume rendering

Right Panel

Displays

- Integral value
- Area
- Volume
- Mass
- Moments
- Center of mass coordinates

Bottom

Buttons

- Rotate
- Zoom
- Animate
- Export
- Save Figure

FINAL DELIVERABLES

- MATLAB App Designer project (.mlapp) providing the primary graphical interface.
- MATLAB source files organized into reusable modules for integration, visualization, and engineering calculations.
- A README explaining the mathematical concepts, software architecture, GUI functionality, and instructions for running the application.
- Sample demonstrations illustrating double integrals, area calculations, mass properties, polar-coordinate integration, triple

integrals, center-of-mass calculations, and engineering case studies.

multivariable calculus

$$\iiint f(x,y,z) \frac{\partial f}{\partial y} \frac{\partial f}{\partial x} 7i + 4j + 3z$$